Term Project

CS 151: Object Oriented Design

Technical-Spec-v1

Prof: Ahmad Yazdankhah

Team Number	19
Software Name	KnowledgeTrack

#	Name	Major	Accomplishment
1	Sean Reece Calantoc	CS	Clean-Up/Intro
2	Kanishka Yadav	DS	UML Class Diagram
3	Hoang Khang Pham	SE	UML Sequence Diagrams
4	Cheyenne Khouri	CS	Overview, Data model

1 Introduction

1.1 Objective

This document is the technical description of the software called KnowledgeTrack to help faculty members have easy access to all the information of each student. This software allows faculty members to search for students either by name or any other characteristic, such as academic status, job status, programming language knowledge, etc. As well as having the ability to edit, create, or delete profiles.

1.2 References

1. Yazdankhah, Ahmad “CS151-L07-P1-GUI Programming”

2. CS151 lecture notes on data modeling and UML (class materials).

3. diagrams.net (draw.io) — used to create UML diagrams.

1.3 Acronyms and Abbreviations

UML	Unified Modeling Language
CSV	Comma-Separated Values
GUI	Graphical User Interface

2 Software Overview

2.1 Functional Requirements

1. Faculty can look for a student by searching their name
2. If there are multiple students with the same name, a list is provided for the user to choose from
3. If a faculty member can’t find a student through their name, they can use achievements, academic status, or career goals to find them
4. Once the student is selected, a list of achievements, academic status, skills, career goals, and any other information is listed
    a. As well as information such as a current job, job details, programming languages they know
5. There are options to edit or delete the profile

    a. There is a textbox if faculty wishes to add comments over time
6. A checkbox is provided to choose if a student will be whitelisted or blacklisted
7. If a faculty member tries to create a profile for a student under the same name, an error will be shown
8. Faculty can also generate reports in bulk (ex. Listing all students that share a common skill, all students on blacklist or whitelist) or just generate the report of a singular student

3 Detailed Design

3.1 Data Model

3.1.1 Data Storage Characteristics

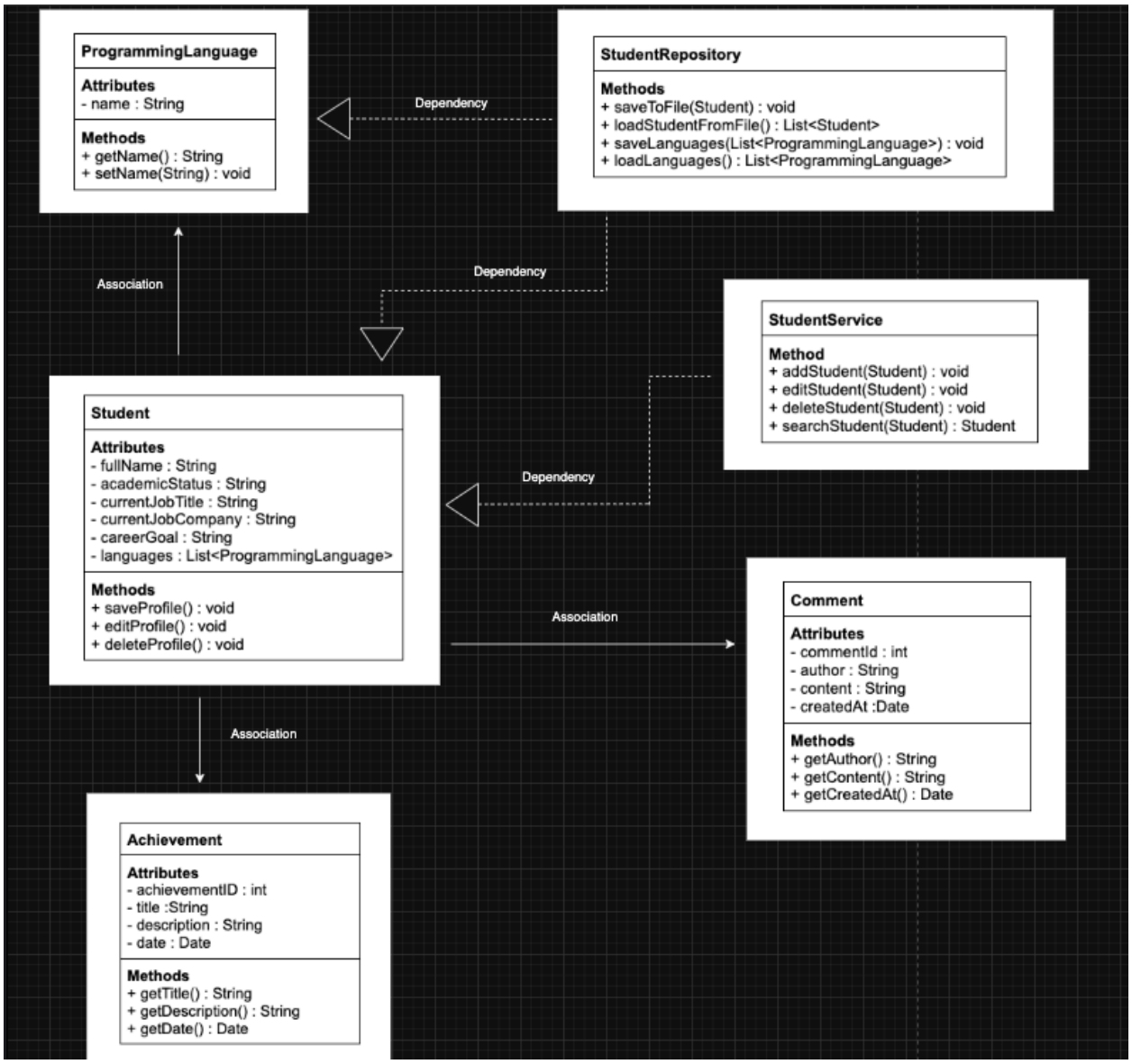
Storage type	Flat File

3.1.2 Entities Name

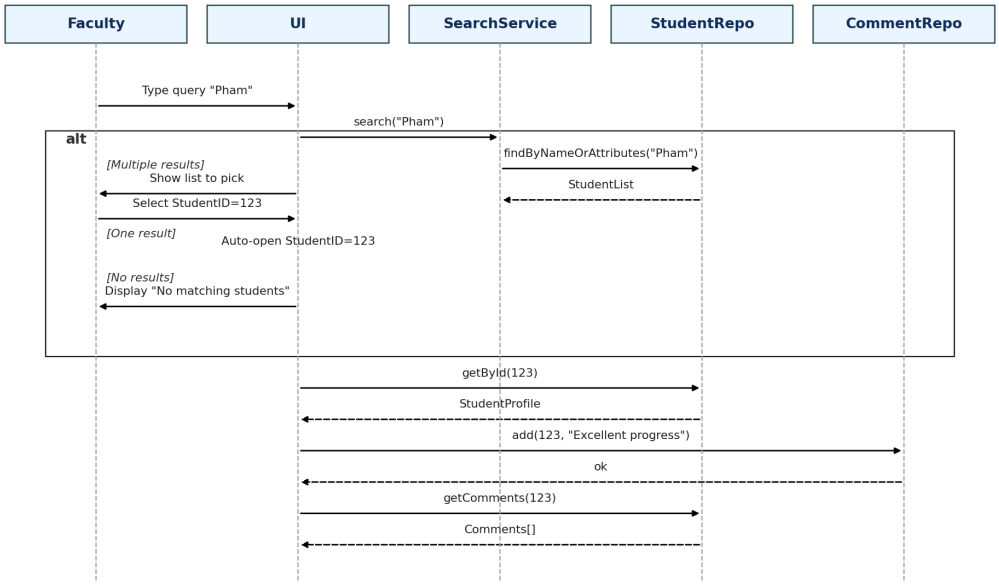
1.	Language.csv
2.	Student_Profile.csv
3.	Student_Language.csv (join for Student <-> Language)
4.	Skill.csv
5.	Student_Skill.csv (join for Student <-> Skill)

6.	Achievement.csv
7.	Comment.csv

3.2 UML Class Diagrams



3.3 UML Sequence Diagrams



Sequence: Search student → open profile → add comment → refresh comments